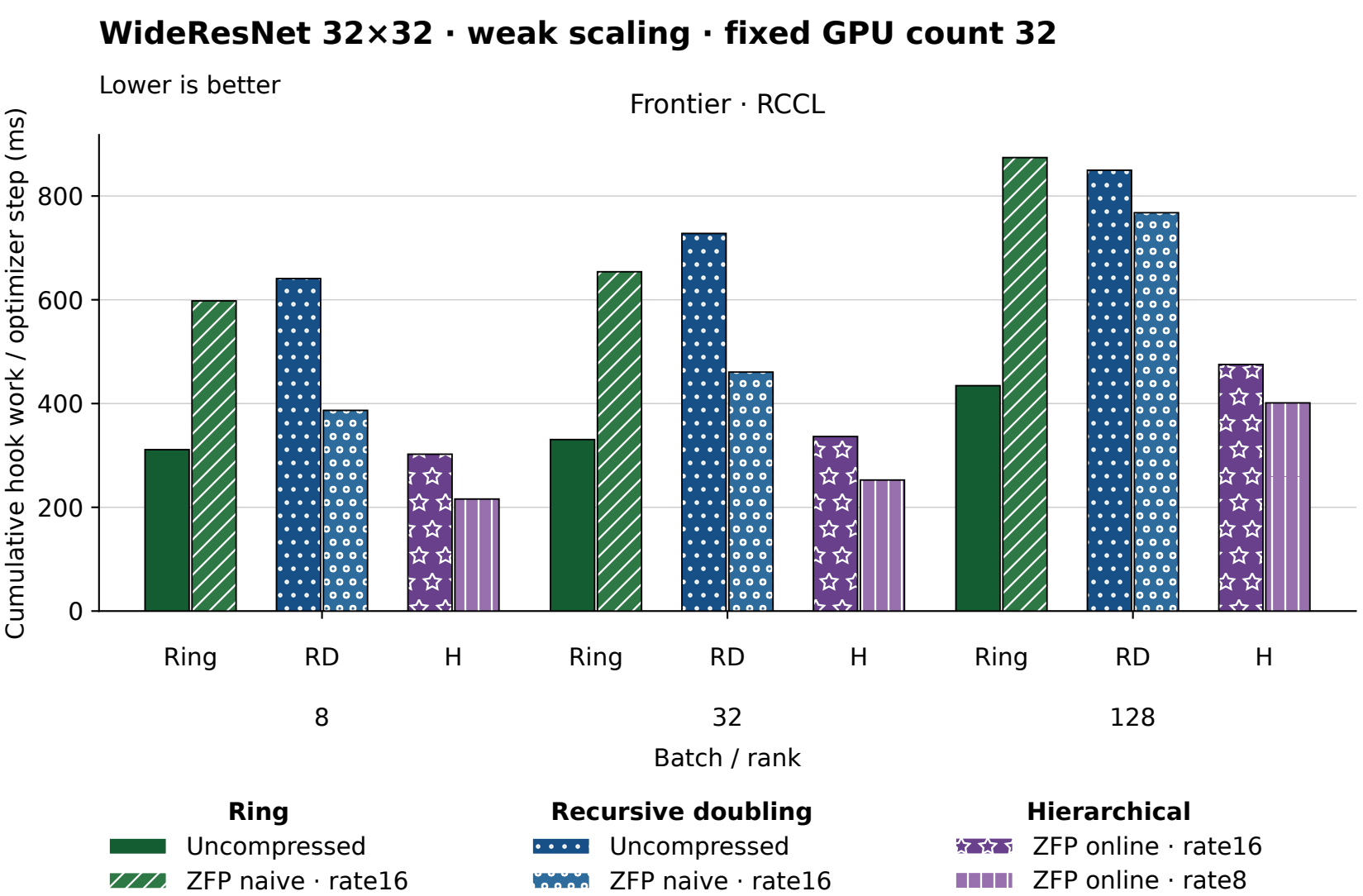

WideResNet 32×32 · weak scaling · fixed GPU count 32
Lower is better
Frontier · RCCL
Cumulative hook work / optimizer step (ms)
0
200
400
600
800
Ring
RD
H
Ring
RD
H
Ring
RD
H
8
32
128
Batch / rank
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate